Mohana Krishnan M V

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Objective

B.Tech Computer Science student (Expected May 2027) with a strong foundation in **Python** and **Object-Oriented Design**, and working knowledge of **Java** and **C**. Demonstrated experience delivering **AI/ML research projects**, production-grade backend systems, and **UI/UX prototypes** from concept to deployment. Seeking to apply **Machine Learning** and **Software Engineering** expertise to build impactful, real-world solutions.

Technical Skills

Languages	:	Python, TypeScript, Java, C, SQL, Dart, HTML/CSS
Backend & Data	:	NestJS, Node.js, FastAPI, Socket.io, PostgreSQL (PostGIS), Redis, Supabase
Libraries & Frameworks	:	React, TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Flutter
Developer Tools	:	Docker, Git, GitHub, VSCode, Webots, Jupyter Notebook, Google Colab

Education

Amrita Vishwa Vidyapeetham — Expected May 2027

Bachelor of Technology in Computer Science — *Coimbatore, India*

- **CGPA:** 7.12/10
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Operating Systems, Database Management Systems, Computer Networks

Experience

KVT Exports — Jan 2026 – Mar 2026

AI/ML Engineer Intern — *Coimbatore, India*

- Designed a **product recommendation system** combining **2 filtering strategies** — collaborative and content-based — to personalize customer product discovery at scale.
- Developed an **LLM-driven auto-reply engine** (GitHub) that autonomously drafts contextual support responses from incoming product queries, reducing manual response time by over **60%**.
- Unified **ML inference outputs** and **GenAI workflows** within the front-end interface, streamlining support flows and improving end-to-end response consistency.

Projects

Vectra: Intelligent Ride-Sharing & Pooling Platform — *NestJS | Flutter | PostGIS | Redis*

- Built a production-grade **microservices system** spanning **2 platforms** (NestJS backend + Flutter mobile), with **Socket.io** powering real-time driver tracking and GPS synchronization.
- Engineered a custom **ride-pooling algorithm** grouping compatible riders via **detour penalty scoring** and multi-factor compatibility matching to maximize route efficiency.
- Established a geospatial data layer using **PostgreSQL + PostGIS** for efficient **nearby-driver discovery** and radius-based spatial queries.
- Architected a robust **trip state machine** managing complex ride lifecycles with strict **transition validation** and programmatic resource cleanup.

Logistics AI Compliance Checker — *Computer Vision | NLP | Python*

- Secured a **Top Finalist** position at **IIT Bombay's Logithon 2025** (national-level competition) by building an end-to-end AI system for **logistics compliance verification**.
- Fused **Computer Vision (OCR)** and **NLP** pipelines to extract structured, queryable data from unstructured shipping manifests with high reliability.
- Constructed a **rule-based validation engine** that cross-references extracted data against regulatory standards, significantly reducing manual audit time and human error.

Gaze Prediction via Fractional-Order Modeling — *Ongoing Research | Python | Signal Processing*

- Designing a multi-stage **Python pipeline** to predict future gaze positions by clustering **eye-tracking data** and identifying behavioral patterns across user sessions.
- Applying **fractional calculus** to model non-local memory effects in human visual attention, targeting higher prediction accuracy over conventional integer-order baselines.
- Preprocessing and normalizing large-scale **coordinate datasets** to strengthen model robustness and generalization across diverse user behaviors.

Ontology-Guided Floor Plan Generator *GenAI | Knowledge Graph | Constraint Solving*

- Represented architectural domain knowledge through an **ontology** and **knowledge graph**, enforcing spatial design rules and room relationship constraints systematically.
- Leveraged a **constraint-solving approach** for geometric layout generation, producing **multiple valid floor plan configurations** per input specification.
- Formulated **ranking metrics** — including privacy score and spatial compactness — to objectively evaluate and compare generated layouts across configurations.

Federated Learning Simulation for Edge Networks *Python | Distributed ML | Academic Project*

- Directed a **5-member team** to re-implement and evaluate a **client selection strategy** for clustered federated learning in wireless edge network environments.
- Performed simulation-based analysis of **distributed learning behavior** using Python and computational intelligence techniques to assess model convergence and fairness.
- Oversaw full project lifecycle — from research and implementation through simulation analysis and final reporting — ensuring timely delivery of all milestones.

Vision-Based Glacier Density Monitoring *Independent Research | Deep Learning | Remote Sensing*

- Developed a **digital image processing** pipeline to analyze temporal variation in the Siachen Glacier, enabling data-driven **climate risk assessment** from satellite imagery.
- Compared **deep learning** classifiers against traditional image processing baselines, achieving a peak accuracy of **93.42%** on real-world environmental data.
- Collected, curated, and annotated large-scale **satellite imagery datasets** to support robust model training and evaluation across multiple glacier states.